Bio-inspired Robotic Fish Enabled Oil Spillage Monitoring

Umar Masood, Navid Khiabani, Kianoosh Karimi, Wenyu Zuo, Zheng Chen, Haleh Ardebili, Rafael Verduzco

***Abstract*—This paper presents classification of oil spillage sensor based on organic electrochemical transistor (OECT) technology, and its integration with a bio-inspired robotic fish platform for autonomous oil detection. The OECT sensor is designed to detect crude oil in aqueous environments, including seawater, with high sensitivity and stability. The sensor's signal conditioning circuit is developed to enable real-time monitoring and classification of oil presence. The AquaNavigator robotic fish platform is designed to carry the OECT sensor, providing mobility and adaptability for various underwater applications. Lab testing of the OECT sensor demonstrates its capability to detect oil concentrations as low as 10µg, while the signal conditioning circuit allows for efficient data acquisition. The AquaNavigator platform is equipped with a modular design, enabling easy integration of different sensors and payloads. Experimental results are performed in an indoor water pool facility that show the effectiveness of the OECT sensor and the AquaNavigator platform in detecting oil spills, shows the feasibility of using such sensor for the detection of oil spillage source in water bodies. The proposed system has potential applications in environmental monitoring, underwater exploration, and search and rescue operations.**

***Index Terms*—OECT, Robotic Fish, Bio-inspired, Oil Sensing**

I. Introduction

UNTIL 2017, around 37% of the global crude oil extraction was being done through the off-shore drilling [?]. The crude oil being very useful for the energy production, is also a major source of pollution in the oceans. The oil spills can be caused by various reasons, such as accidents, leaks, or natural disasters. These spills can have devastating effects on marine life [?], ecosystems [?], and coastal communities [?]. The detection and monitoring of oil spills is crucial for mitigating their impact and preventing further damage. Oil leakage detection techniques are generally classified into internal (also known as software-based) and external (hardware-based) approaches. The internal approach is based on computational methods, while external techniques typically rely on local/remote sensing, with further division into sensing/detection devices and visual or surveillance systems subcategories[?].

The most common internal techniques are mass-volume balance [?] , negative pressure wave (NPWs) [?] , pressure point analysis (PPA) [?] and dynamic modeling [?]. These computational methods utilize internal fluid measurement instruments to monitor key parameters along the pipeline [?]. Discrepancies in the measured values between the inlet and outlet of the pipeline are analyzed to determine the presence and location of a leak. While internal methods are suitable for real-time monitoring and can effectively detect and localize leaks, they often involve high implementation costs and are vulnerable to measurement noise, which may lead to false alarms [?].

External leak detection methods are generally more cost-effective than internal techniques and offer higher accuracy in localizing the leak point. The first subcategory of these methods involves the utilization of sensors to monitor the external environment of the pipeline, enabling the detection of anomalies and potential leakage events [?]. Regardless of the specific sensing principles, these methods typically require contact between the monitoring infrastructure and the sensing element such as acoustic[?] , fiber optic[?] , fluorometry[?] and capacitive [?] based sensors.

All the above-mentioned sensors work based on detecting changes in either environmental or physical parameters surrounding the pipeline to localize the leak. However, depending on the specific application certain sensors may be more practical or suitable than others. For instance, to monitor long subsea pipeline segments, acoustic and fiber optic sensors are widely adopted due to their high sensitivity and capability; however, their deployment can be costly and technically complex. Capacitive and fluorescence-based methods are comparatively low-cost and effective in controlled environments but tend to be less reliable under harsh subsea conditions [?]. The second subcategory of external methods involves visual observation, employing optical cameras, and video sensors. These systems can offer wide sensing coverage and effective leak identification with low false alarm rate when visibility conditions are favorable, however they are not suitable for continuous, real-time monitoring due to operational and logistical limitations. Recent advancements of remotely operated vehicles (ROVs)[?] have remarkably enhanced the surveillance capabilities of offshore oil transport infrastructure, with further evolution into autonomous underwater vehicles (AUVs) [?] , enabling more autonomous and efficient operations. Aerial and satellite imaging also fall within this category, as effective tools for detecting large-scale oil spills; however, as mentioned before they suffer from delayed response times relative to the moment of leak occurrence. Considering the pros and cons of each approach, the most effective leak detection strategies for subsea pipelines seem to be the ones that combine direct sensing technologies with ROV/AUV-based inspections [?].

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Carrying the sensors on a robot is a common practice, specially for the environments where a human operation is not safe or possible. Not only underwater, but in the ground robots and drones as well [?]. For underwater robots, the remotely operated vehicles (ROVs) are the most common platforms, which are used for inspection and monitoring of underwater structures. Qasem et al. [?] presented a multisensory ROV platform capable of detecting oil spills through the camera surveillance and image processing techniques. Yao et al. [?] developed an ROV equipped thin oil slick sensing system based on a generalization of the sparce iterative covariance estimation (SPICE) to calculate time of flight of the ultrasonic signal. There are other ROV systems developed for oil spill detection based on acoustic and optical imagary [?], [?], but to the best of our knowledge, there is no robotic systems developed for oil spill detection based on chemical sensors.

However, these ROVs are often expensive and require specialized training to operate. Moreover, these ROVs bring a lot of turbulence in water naturally caused by the propulsion method they employ [?]. In contrast, a bio-inspired robotic fish are emerging in this domain, because of their inherent ability to swim smoothly in water, and minimally affecting the aquatic eco-system [?], [?]. These robotic fish are designed to mimic the swimming motion of real fish, which allows them to move through water with minimal resistance and turbulence. This makes them ideal for applications such as underwater inspection, monitoring, and exploration. In the proposed work, we present a bio-inspired robotic fish platform that is capable of carrying the OECT sensor for oil spill detection. The platform is designed to be modular, allowing for easy integration of different sensors and payloads. This modularity enables the robotic fish to adapt to various applications, such as environmental monitoring, underwater exploration, and search and rescue operations. The proposed platform is also designed to be cost-effective and easy to operate, making it accessible for a wide range of users.

The rest of the paper is organized as follows: Section II describes the OECT sensor and its signal conditioning circuit design. Section III presents the design and fabrication of the AquaNavigator robotic fish platform. Section IV discusses the testing and validation of the signal conditioning circuit in the lab setup. Section V presents the experimental results of the OECT sensor and the AquaNavigator platform in the water pool facility. Finally, Section VI concludes the paper and discusses future work.

II. OECT Sensor for Oil Detection

The oil detection sensor used in this work is an organic electrochemical transistor (OECT) developed by [?]. This thin-film device is fabricated using a PEDOT:PSS channel coated with a hydrophobic polystyrene layer to enable selective adsorption of crude oil. When oil contacts the surface, it alters the ionic permeability of the channel, increasing impedance and reducing the source-drain current. The device operates in aqueous environments, including seawater, and is sensitive to oil concentrations as low as 10µg. It maintains stability over extended immersion periods and can be fabricated on flexible substrates. These properties make it suitable for integration into mobile robotic systems for in-situ monitoring.

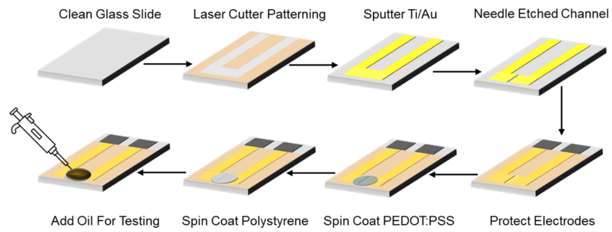
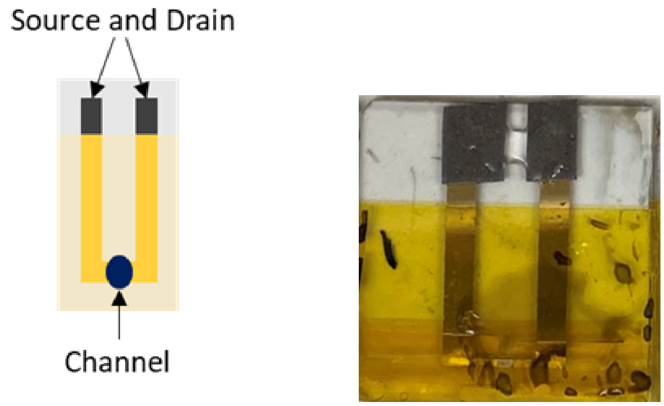


Fig. 1: OECT sensor for oil detection. The sensor is fabricated on a glass substrate with gold electrodes and a PEDOT:PSS channel coated with a hydrophobic polystyrene layer.



(a) OECT sensor schematic

(b) OECT sensor fabricated on glass substrate

Fig. 2: OECT sensor: schematic and fabricated version

A. *OECT Sensor Fabrication*

The organic electrochemical transistors (OECTs) were fabricated on 1.0×2.5 cm glass substrates, where Kapton tape was used to define the electrode configuration before depositing 5 nm titanium and 50 nm gold layers via sputtering. Following electrode protection with Kapton tape, the channel was spin-coated with a PEDOT:PSS mixture containing (3-glycidyloxypropyl) trimethoxysilane (1 wt%), ethylene glycol (5 vol%), and 4-dodecylbenzenesulfonic acid (0.25 vol%) at 1000 rpm for 30 seconds, followed by thermal treatment at 140 °C for 1 hour. The channel was then patterned by mechanically etching the metal layers with a syringe needle and subsequently coated with PEDOT:PSS and an additional polystyrene (PS) layer. The hydrophobic PS enabled oil adsorption upon contact, modulating the ionic permeability of the channel and thereby influencing the source-drain current of the OECTs [?].

The transfer characterisitcs of the OECT was evaluated using a source meter. Complementary electrochemical characterization was further conducted to observe the changes in resistance and ionic permeability using cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS). Fluorescence microscopy was used to visualize the adsorption of

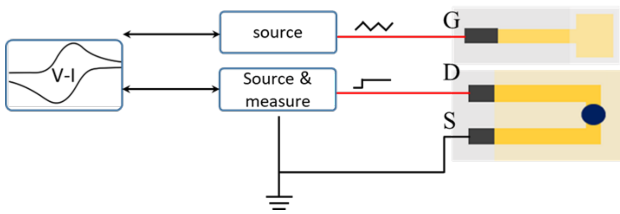


Fig. 3: Schematic of the lab testing setup for OECT sensor characterisation.

crude oil on coated surfaces. Contact angle measurements were implemented to assess oil interaction with the sensing surface. This method showed improved oil spreading in coated samples compared to non-coated ones in a previous study [?].

B. Lab Testing of the Sensor for Classification

Using an OECT to detect crude oil has several advantages: It has a low fabrication cost and is stable in aqueous environments. Its transmitter design also allows the signal to be easily amplified and read. These advantages ensure rapid detection and easy deployment when integrated into robots. Based on OECT characterisation, a gate voltage, -0.6V to 0.6V, is selected to obtain measurable current and limited water electrolysis. Meanwhile, a constant source-drain voltage of -0.6V is applied to the source-drain. Considering the results obtained from normal water as the baseline. The transfer characteristics of each of the OECT sensors were determined by measuring the drain current over a range of gate voltage before and after the addition of crude oil.

From the previous work [?], the lab test observed that adding oil can inhibit electron transfer over the channel, thus increasing impedance and resistance. This leads to a decrease in current in the transfer characteristics. The lab result is obtained through cyclic voltammetry, which employs a linearly sweeping signal at the electrode in both directions. In this work, a sawtooth input voltage has been applied to the gate to collect data for a plot of current versus potential. A source measurement unit (SMU) has been utilized to monitor the current flowing between the source and drain, while maintaining a constant voltage. The equipment use in laboratory is Keithley Digital Multimeter 2100 for gate voltage, and Keithley's Series2400 Source Measure Unit for drain-source voltage and current measurement.

C. Signal Conditioning and Circuit Design

While the lab setup allowed for precise characterization of the sensor's response to oil, it was not practical for real-time or mobile deployment. The source meter is heavy, power-intensive, and not suitable for integration with autonomous platforms like robotic fish. To bridge this gap, a compact and dedicated signal conditioning circuit has to be developed. This circuit needed to replicate the waveform generation capabilities of the lab setup—such as applying sawtooth or square wave voltages—while simultaneously measuring low currents with high resolution. The design included three key channels: a variable gate voltage output, a constant drain voltage supply, and a current-sensing input at the source. Effective signal conditioning ensures clean acquisition of the sensor's output, enabling downstream classification logic to accurately detect oil presence during robotic operations in field conditions.

For this purpose a circuit is immensely needed which can perform the similar tasks initially done by the SMU's with on-board data acquisition, and has to be small enough to be put up of light weight robot. The primary function of the circuit is to (1) supply constant voltage to the source of the OECT channel, (2) supply +0.6V to -0.6V to the gate, (3) measure the current flowing from the source to the drain, (4) provide a means for data communication with the robotic fish and (5) be able to communicate with the host computer using UART interface. INA219 chip from Texas Instruments is used to sense current, with a resolution as high as $10\mu A$, when used with a shunt resistor of value 110Ω. For each of the voltage supply, MCP4725 chip from Microchip Technology is used which can receive the command from the main microcontroller on Inter-Integrated Circuit (I2C) bus. The microcontroller used for this purpose is ESP32-C3 from Espressif Systems.

A block diagram illustrating the functionality of the circuit is shown in Fig. **??**. The details of the circuit and the full schematic are provided in the Supplementary Information section.

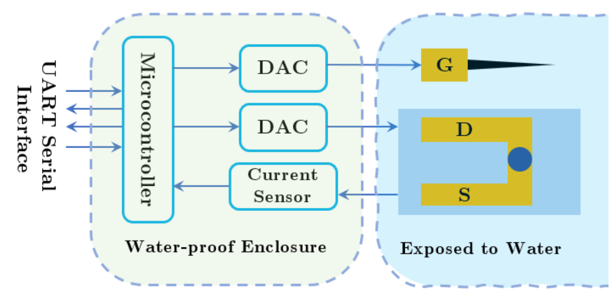


Fig. 4: Block diagram of the OECT signal conditioning circuit.

D. Enclosure and Hardware

To be able to use this circuit onto an underwater robot, a waterproof enclosure is also designed. The enclosure is designed as a waterproof chamber with two waterproof cord ports - one for power and data transfer and the other for housing the gate. The circuit sits at the bottom of the chamber. In the middle section of the enclosure, there is a cap with an o-ring that seals the enclosure. The cap also has a wire pathway for the cable and an anode for the drain-source pair. The sensor must be placed downward at the cap, with its drain-source touching the anode and cathode. An additional layer of sealing form made of silicon rubber is placed around the sensor. A sealing cap covers and compresses the sealing form, stopping water from touching the sensor. All the enclosures and caps are connected by screws.

III. AquaNavigator - The sensor carrier

To enable autonomous and mobile oil spill detection, we integrated the OECT sensing system into a custom-designed

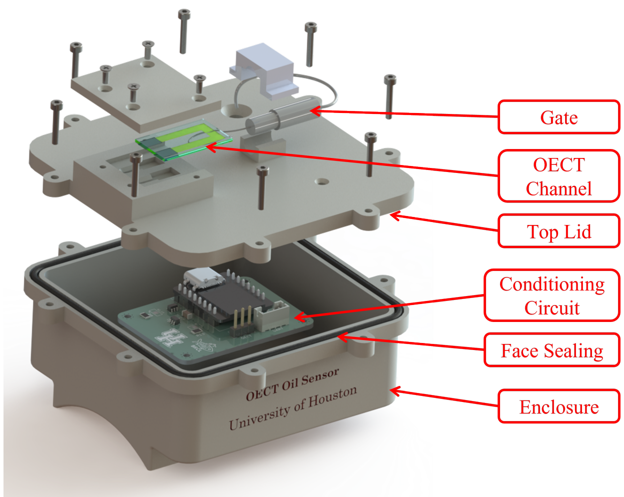
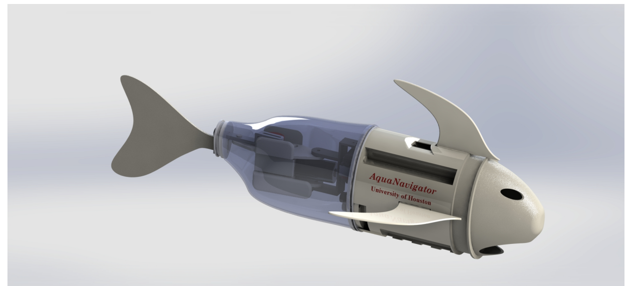


Fig. 5: Enclosure for the OECT sensor and signal conditioning circuit

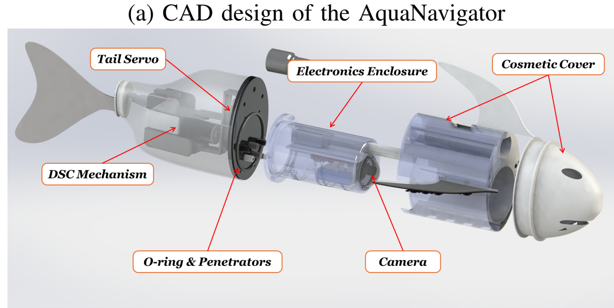
robotic fish platform, referred to as the AquaNavigator. This underwater vehicle is designed to operate in both static and dynamic conditions, allowing it to patrol confined tanks or trace pipelines in open water environments. The AquaNavigator serves as the physical carrier for the sensing circuit, providing power, mobility, and data collection capabilities. Its hydrodynamic form factor, propulsion system, and onboard electronics are tailored for smooth underwater operation while minimizing sensor interference. By embedding the oil detection system within this mobile unit, we aim to expand the application of OECT sensors from lab-based trials to real-world aquatic environments. The following subsections describe the mechanical design, fabrication process, waterproofing strategy, and integration of sensing and control modules.

A. Design of AquaNavigator

AquaNavigator is designed to be a modular platform developed previously by our group [**?**], keeping efficiency in mind, thanks to the double slider-crank mechanism [**?**]. The robotic fish contains a buoyancy based depth control system [**?**], which adds third degree of freedom. The design and shape of the AquaNavigator is shown in Fig. **??**, the main parts include, the tail assembly, the BCD and electronics enclosure, and the top cosmetic cover. The tail assembly consists of a DSC mechanism operated by the DC motor, and the tail-bias angle control using the servo motor. All the parts in the tail assembly are water-proofed using epoxy resin at the component level. The BCD and electronics enclosure is designed to house the electronics, battery, and camera. The enclosure is waterproofed using face-sealing on the flanged face of the enclosure. The top cosmetic cover is designed to provide a sleek appearance along with the hydrodynamic shape. All the parts are fabricated using Stereolithography (SLA) 3D printers by FormLabs™. The robotic fish is powered by a 3-Cell Lithium Polymer (LiPo) battery, which provides a voltage of 11.1V and a capacity of 1500mAh, which allows a continuous operation of 60 mins in the water tank. The main control boards is equipped with a 32-bit ESP-32-C3 microcontroller and motor driver ICs to control the motors. All the low level operations are handled by this on-board microcontroller, which also communicates with the host computer using the wireless communication or the wired Serial communication, depending upon the application.



(a) CAD design of the AquaNavigator



(b) Exploded view showing main components

Fig. 6: Design of AquaNavigator robotic fish platform.

B. Communication and GUI on Host Computer

The AquaNavigator is capable of wireless communication (in shallow water only) and wired communication with the host computer. In different applications, the communication can be used to send the sensor data to the host computer for further processing, or to control the robotic fish. For the purpose of this study, to show the proof-of-concept, the robotic fish is tasked to track an oil pipeline in the water tank. The robotic fish has on-board RGB camera, which can be used to track the pipeline using Visual Servoing technique as shown in one of the previous work of the authors [**?**]. For this application, most of the computation is performed in the host computer running a Windows program developed in Python. The host computer is connected to the robotic fish using a USB cable, which provides a reliable and fast communication link. The program performs all the necessary computations, including but not limited to; (1) image acquisition, (2) image processing and hough transform to detect the pipeline, (3) control signal generation based on the visual servoing algorithm, (4) communicating the control signal to the robotic fish, and (5) showing the real-time video feed and control signals on the GUI. The GUI contains user controls to control the robotic fish in the manual operation mode, or set the parameters for the visual servoing algorithm in the autonomous operation mode. The GUI also displays the real-time video feed from the on-board camera, which can be used to monitor the operation of the robotic fish. The GUI is shown in Fig. **??**.

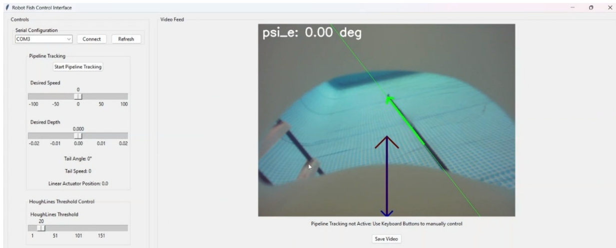


Fig. 7: Graphical User Interface (GUI) for controlling the AquaNavigator and visualizing the sensor data.

IV. Experimental Validation and Results

A. Fabrication and Integration with AquaNavigator

The AquaNavigator is capable of the carrying a sensor in its sensor housing, and the main control board allows to connect one sensor with a Universal Asynchronous Receiver-Transmitter (UART), Serial Peripheral Interface (SPI), or Inter-Integrated Circuit (I2C) interface. The OECT sensor is designed with UART interface, and can be wired to the main control board of the AquaNavigator. The OECT sensor can be physical mounted on the top of the robotic fish to have maximum exposure to oil, considering the oil accumulate on the surface of the water. This complete integration is shown in Fig. **??**. The OECT transducer has to be submerged in the water to sense the oil, but the signal conditioning electronics which is an essential part of the sensor, has to be waterproofed. For this purpose, the enclosure for the circuit is fabricated using SLA 3D printing.

B. Experimental Setup

The experimental study in this research consists of two parts: (1) lab testing for the characterization of the OECT sensor, which would later guide the design of the signal conditioning circuit, and (2) proof-of-concept testing of the OECT sensor integrated with the AquaNavigator robotic fish in an indoor water pool.

1) Lab Testing Setup: The lab testing setup is designed to characterize the OECT sensor and its response to different water solutions. The setup consists of a Source Measure Unit (SMU) connected to the OECT sensor, which allows for precise control of the gate voltage and measurement of the source-drain current. The SMU is programmed to apply a sawtooth waveform to the gate of the OECT sensor, while maintaining a constant voltage at the drain. The current response of the sensor is recorded over time, allowing for analysis of its sensitivity and stability in different aqueous environments. The second part of the lab testing is the testing with the designed signal conditioning circuit. For this one, the OECT is assembled into the enclosure Fig. **??**, and through the USB cable, the circuit is connected to the host computer, running a MATLAB script to visualize the sensor data in real-time. The MATLAB script is designed to read the gate voltage and source-drain current from the signal conditioning circuit, and plot the current response of the OECT sensor over time.

figures/watertank-top.png

Fig. 8: The top view of the water tank setup for OECT testing.

2) Water Tank Testing Setup: For the proof of concept testing of the OECT sensor integrated with the AquaNavigator robotic fish, a water tank is used. The water tank is big enough that a normal oil leakage flow and absorption can be achieved. The tank measures 10 m long, 5 m wide and 1.5 m deep. The tank is filled with fresh water, and a black colored pipeline of oil is placed at the bottom of the tank. The pipeline is connected to a pump, which is used to circulate the oil in the pipeline. The oil is circulated at a constant flow rate, which is controlled by a valve. At a certain point in the pipeline, a small hole is made to allow the oil to leak out of the pipeline. This setup allows to simulate the oil leakage in a controlled environment, and the OECT sensor can be used to detect the oil leakage.
Next, the OECT sensor is integrated with the AquaNavigator robotic fish, and the fish is programmed to swim along the pipeline. The OECT sensor signal conditioning circuit is designed with a Serial Universal Asynchronous Receiver-Transmitter (UART) interface, which allows it to communicate with the main control board of the AquaNavigator. But for the purpose of experimental testing, and be able to visualize and record the sensor data in real-time, the OECT sensor is connected to a host-computer through UART interface communication. The host computer is running the same MATLAB script as used in the lab testing.

C. Lab results of OECT Characterization

The figure **??** (Multi-solution test) illustrates the time response of the OECT at various liquid samples, including water, soda, seawater, and oils. The input is sawtooth gate voltage, and the current response exhibits a triangle wave format and shows distinct electrical behaviors across the solutions. In general, the non-oil sample indicates high conductivity at the OECT channel due to its ionic content, and the oil samples display significantly lower current levels, less than 0.2 mA,

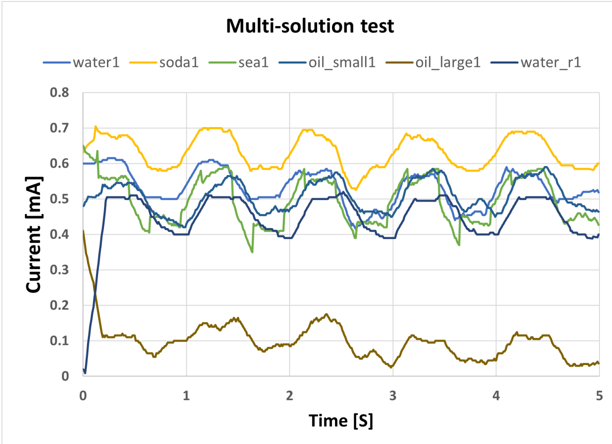


Fig. 9: The time response of the OECT sensor in different liquid samples.

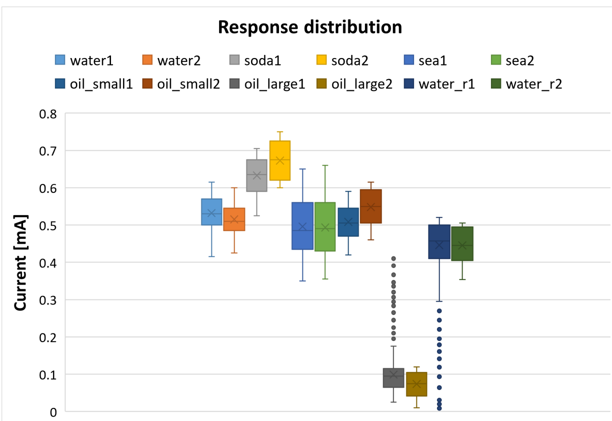


Fig. 10: The normalized current response of the OECT sensor in various liquid samples.

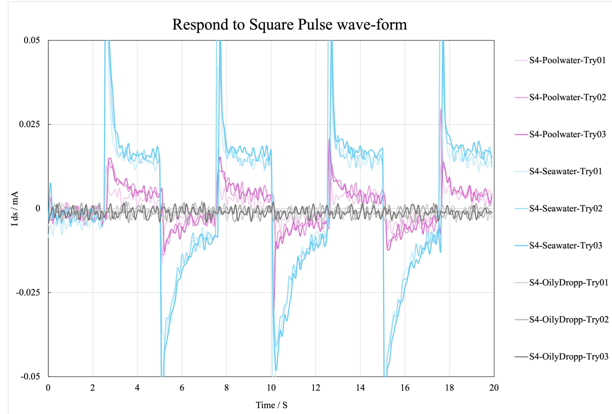


Fig. 11: The response of the sensor while it is tested with the circuit setup for current measurement instead of source measurement unit and oil droplet is injected directly on the channel of the OECT before being immersed into water.

reflecting the poor conductivity at the OECT channel and indicating the contacting of the oil molecule. The periodic fluctuations observed in every sample prove the dynamics between input signals and output current strength, which can be denoted as a hysteresis. The figure **??** (Response distribution) uses a box plot to compare the current strength across various liquid samples, clearly illustrating the current drop when the OECT contacts the oil solution. Similar results are provided in as the previous figure; the current response is still consistent in non-oil samples, while the oil sample leads to a significant current drop. The lab testing proves that the OECT is an effective sensing method to detect if an oil molecule exists in the liquid sample, and the sawtooth excitation is feasible to obtain sensing result similar to cyclic voltammetry,

The characterization of the OECT was performed in an experimental setup in which the same water that will be used in a research pool to test the robot is used and compared to sea water. Figure **??** shows the sensor response to the square wave generated by the circuit setup illustrated in Figure **??**. The experiment was first performed while the sensor was immersed in the pool water, and the experiment was repeated three times. The purple diagrams in figure **??** show the sensor response in this stage of the experiment. These three diagrams show good repeatability in the sensor response.

Subsequently, the experiment was repeated with sea water taken from the Galveston beach in Houston. The experiment was again repeated three times and the blue diagrams of figure **??** were recorded. These three diagrams are again very close to each other showing the sensor has also a good repeatibility in its response to squarewave in sea water. The current of the sensor in sea water is more than two times larger than that of pool water which is due to higher salt concentration in sea water. This shows that the sensor can clearly show the difference in the conductivity of the medium.

As the third step, one dropplet of crude oil $(10\mu L)$ was injected onto the channel of the OECT, and then it was immersed into sea water, again repeated for three times. In all of the three experiments of this stage, the sensor response to gate excitation was almost close to zero and showed that the sensor lost its functionality as a transistor when crude oil was injected directly on its channel. This behavior shows that the sensor can clearly show the presense of crude oil by not responding to gate voltage.

Injecting the crude oil on the channel of the OECT before immersing it into water is not enough similar to real conditions in which the robofish is expected to detect oil leackage in sea water. In another experiment, two different amounts of crude oil were added to sea water and the response was recorded and compared to that of sea water and pool water. Figure **??** illustrates the waveforms of the sensor response to square wave exerted to the gate with two different concentrations of crude oil. Each experiment was repeated three times, and the results prove the repeatability of each step. By comparing the diagrams for the two concentrations of crude oil, it can be observed that the conductivity decreases by inceasing the crude oil concentration. These results not only show that the sensor can detect the crude oil, but also it can distinguish oil concentration if characterized well.

Fig. 12: The response of the sensor while it is tested with the circuit setup for current measurement instead of source measurement unit while oil is added to water with concentrations of 4,and 8 g/L.

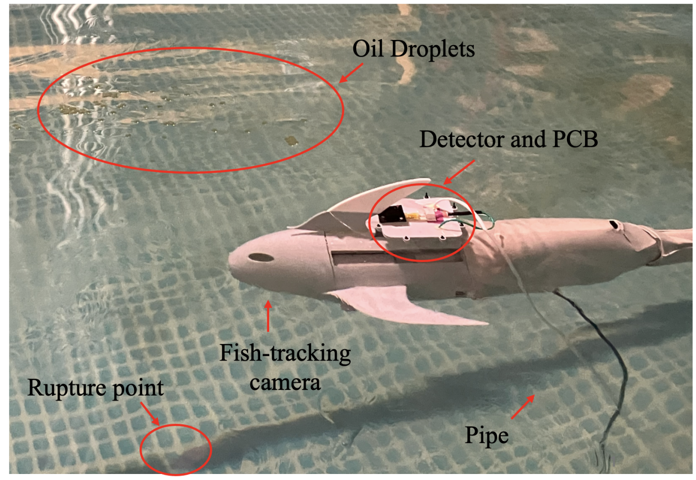


Fig. 13: Experimental setup for testing the robofish in the pool.

D. Water Tank results of OECT with robotic fish

The AquaNavigator, an intelligent underactuated robotic fish, serves as the carrier for the OECT sensor, enabling dynamic and adaptable monitoring of oil spills in aquatic environments. To validate the sensor's performance in practical conditions, the robotic fish is programmed to autonomously track a pipeline in a water tank using an image-based visual servoing technique. The on-board camera provides a video feed to the host computer, where the pipeline is detected and control signals are generated and transmitted to the fish's main control board for motor actuation. As established in our previous work [**?**], two key parameters are used to characterize the tracking performance: the heading error angle ψ_e, defined as the angle between the fish's current heading and the desired heading towards the projected pipeline point, and the pixel distance y between the head of the heading vector and the desired pipeline point. In addition to these image-based metrics, the tracking accuracy is also evaluated by measuring the fish's angular deviation from the aimed heading vector and its lateral distance from the pipeline, as illustrated in Fig. **??** and Fig. **??**.

For testing the integrated system, the sensor was mounted on the robofish and tested in a pool where the robofish tracked a pipeline. An image of the robofish in this step of the experiment is shown in figure **??**. The sensor response to a square wave was first recorded while the robofish remained stationary and no oil was present in the pool. This measurement served as the baseline for the experiment. Afterwards, the sensor response was recorded again while the robofish was moving along the pipeline, still in the absence of oil. This step was carried out to identify any influence of the motion of the robofish on the sensor output. Both of these steps were performed three times to fulfill the repeatability of the measurements. The sensor response without oil while the Robofish was stationary compared to the case that the robofish was moving is presented as the light and dark blue curves in Figure **??**.

Once the baseline was established and the effect of movement

Fig. 14: Robotic fish tracking performance along the pipeline.

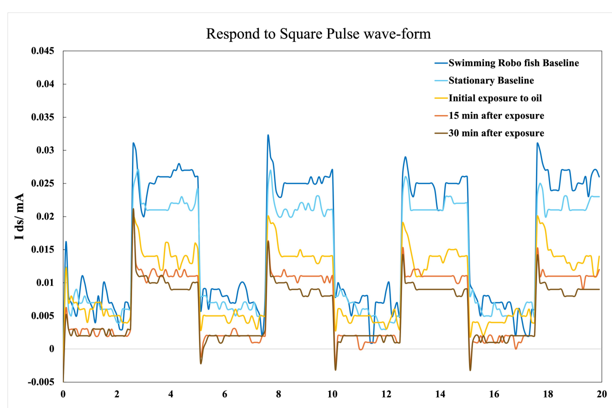


Fig. 15: Response without oil leakage compared with three different exposure levels of the sensor to the contaminated water.

was recognized, an oil leak was introduced at a specific point along the pipeline to determine whether the robofish could detect the leak while passing that location. A constant oil flow rate of 5 mL/min was applied at the rupture point of the pipeline using a peristaltic dosing pump. The yellow curve in figure **??** is the sensor response in the contaminated area. The results confirm that the sensor successfully detected the oil leak when the robofish passed through the contaminated area. After initial detection of the contaminated area, the robofish remains in that area to locate the leakage point. It should be noted that the sensor exhibited a 60-second delay in the amplitude-change after the robofish entered the contaminated zone, which is attributed to the absorption time of oil in the OECT channel. Since there is a continuous oil leakage near the rupture point, the oil concentration continuously increases in the contaminated zone. Therefore, by remaining the robofish in the highly polluted area, the channel of the OECT is expected to absorb more oil, resulting in a larger decline in the observed current amplitude. The measurement was carried out two other times once 15 minutes after the initial detection of the contamination and another time after 30 minutes. The orange and brown curves in Figure **??** show the sensor response after 15 and 30 minutes of exposure to the contaminated zone respectively. This figure shows the reduction in the response of the OECT by higher exposure time to oil leakage.

V. Conclusion

In this paper, we presented a practically feasible solution for oil spillage source sensing using a bio-inspired robotic fish platform. The OECT sensor was designed to detect crude oil in aqueous environments, including seawater, with high sensitivity and stability. The sensor's signal conditioning circuit was developed to enable real-time monitoring and classification of oil presence. The AquaNavigator robotic fish platform was designed to carry the OECT sensor, providing mobility and adaptability for various underwater applications. Lab testing of the OECT sensor demonstrated its capability to detect oil concentrations as low as 10µg, while the signal conditioning circuit allowed for efficient data acquisition. Experimental results in a water tank showed the effectiveness of the OECT sensor and the AquaNavigator platform in detecting oil spills, demonstrating the feasibility of using such a sensor for the detection of oil spillage sources in water bodies. The proposed system has potential applications in environmental monitoring, underwater exploration, and search and rescue operations.

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